

Chapter 13. Geometry in the Plane, Vectors

1. Parametric equations of a plane curve:

$$x = f(t), \quad y = g(t), \quad t \in I$$

- 1) Obtain its Cartesian equation by eliminating the parameter (Page 559,560).
 - 2) Find dy/dx and d^2y/dx^2 without eliminating the parameter (Page 562: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$).
2. Arc-length L of the curve from $t = \alpha$ to $t = \beta$:

$$L = \int_{\alpha}^{\beta} \sqrt{(dx/dt)^2 + (dy/dt)^2} dt$$

3. Sum of two vectors: Triangle Law and Parallelogram Law (Page 567).
4. For any two vectors in the plane $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$, c a scalar,

$$\begin{aligned} \text{vector sum:} \quad & \mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2 \rangle \\ \text{scalar multiple:} \quad & c\mathbf{u} = \langle cu_1, cu_2 \rangle \\ \text{length (magnitude):} \quad & |\mathbf{u}| = \sqrt{u_1^2 + u_2^2} \\ \text{dot product:} \quad & \mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 \end{aligned}$$

5. If $|\mathbf{u}| = 1$, then $\mathbf{u}$ is called a unit (direction) vector.
6. Properties of vector operations: Theorem A in Page 572 and Theorem B in Page 573.
7. When $\mathbf{u}, \mathbf{v}$ are non-zero vectors:

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos \theta$$

where θ is the smallest positive angle between $\mathbf{u}$ and $\mathbf{v}$.

8. Perpendicularity criterion: two vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular (orthogonal) if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.
9. Basis vectors: $\mathbf{i} = \langle 1, 0 \rangle$, $\mathbf{j} = \langle 0, 1 \rangle$. For any vector $\mathbf{u} = \langle u_1, u_2 \rangle$, we have $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$.
10. Vector projection of $\mathbf{u}$ on $\mathbf{v}$:

$$Pr_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}$$

11. Differentiation Formula (Page 578).
12. Curvilinear motion:

$$\begin{aligned} \text{position vector :} \quad & \mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} \\ \text{velocity :} \quad & \mathbf{v}(t) = \mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} \\ \text{speed:} \quad & |\mathbf{v}(t)| = \sqrt{f'(t)^2 + g'(t)^2} \\ \text{acceleration :} \quad & \mathbf{a}(t) = \mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} \end{aligned}$$

Chapter 14. Geometry in Space, Vectors

1. Distance of two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in three-space

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2. Standard equation of a sphere

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

where (h, k, l) is the center and r is the radius (Page 597).

3. Vector in three-space:

$$\mathbf{u} = \langle u_1, u_2, u_3 \rangle = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$$

some operations: $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$,

vector sum: $\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$

length (magnitude): $|\mathbf{u}| = \sqrt{u_1^2 + u_2^2 + u_3^2}$

dot product: $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = |\mathbf{u}||\mathbf{v}| \cos \theta$

cross product: $\mathbf{u} \times \mathbf{v} = \langle u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$

4. Direction angles (Page 600).

5. Standard equation of a plane

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

where $\mathbf{n} = \langle A, B, C \rangle$ is a normal vector to the plane and (x_1, y_1, z_1) is a point on the plane (Page 601).

6. The distance L from the point (x_0, y_0, z_0) to the plane $Ax + By + Cz = D$ is given by

$$L = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

7. Some properties for cross product: given two vectors $\mathbf{u}$ and $\mathbf{v}$ in three-space:

1) $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin \theta$

2) $\mathbf{u}(\mathbf{u} \times \mathbf{v}) = 0 = \mathbf{v}(\mathbf{u} \times \mathbf{v})$

3) $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} \times \mathbf{v} = 0$

4) Theorem C in Page 607

8. A space curve is determined by

$$x = f(t), \quad y = g(t), \quad z = h(t), \quad t \in I$$

$$\begin{aligned}
\text{position vector:} \quad \mathbf{r}(t) &= f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} \\
\text{velocity:} \quad \mathbf{v}(t) &= \mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k} \\
\text{speed:} \quad |\mathbf{v}(t)| &= \sqrt{f'(t)^2 + g'(t)^2 + h'(t)^2} \\
\text{acceleration:} \quad \mathbf{a}(t) &= \mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}
\end{aligned}$$

9. Parametric equations of the line through (x_0, y_0, z_0) and parallel to $\mathbf{v} = \langle a, b, c \rangle$:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

and the corresponding symmetric equations are given by

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

10. The length of arc from $P(a)$ to $P(b)$ is given by

$$s = \int_a^b |\mathbf{r}'(t)| \, dt = \int_a^b \sqrt{f'(t)^2 + g'(t)^2 + h'(t)^2} \, dt$$

11. Some formula for curvature (Page 613-615):

$$\begin{aligned}
\text{unit tangent vector:} \quad \mathbf{T}(t) &= \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \\
\text{curvature:} \quad \kappa(t) &= \left| \frac{d\mathbf{T}}{ds} \right| = \frac{|\mathbf{T}'(t)|}{|\mathbf{v}(t)|} = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3} \\
\text{principal unit normal vector:} \quad \mathbf{N}(t) &= \frac{1}{\kappa} \frac{d\mathbf{T}}{ds} \\
\text{binormal:} \quad \mathbf{B} &= \mathbf{T} \times \mathbf{N}
\end{aligned}$$

12. Quadric surfaces (Page 620-622):

$$\begin{aligned}
\text{ellipsoid:} \quad & \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \\
\text{hyperboloid of one sheet:} \quad & \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \\
\text{hyperboloid of two sheets:} \quad & \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \\
\text{elliptic paraboloid:} \quad & z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \\
\text{hyperbolic paraboloid:} \quad & z = \frac{x^2}{a^2} - \frac{y^2}{b^2} \\
\text{elliptic cone:} \quad & \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0
\end{aligned}$$

13. Cylindrical coordinates (r, θ, z) :

Cylindrical to Cartesian	Cartesian to Cylindrical
$x = r \cos \theta$	$r = \sqrt{x^2 + y^2}$
$y = r \sin \theta$	$\theta = \tan^{-1}(y/x)$
$z = z$	$z = z$

14. Spherical coordinates (ρ, θ, ϕ) :

Spherical to Cartesian

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Cartesian to Spherical

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1}(y/x)$$

$$\phi = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$